

AQUÍ VA EL TÍTULO DEL ARTÍCULO

ABSTRACT. Aquí va el resumen

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1. INTRODUCTION

[PP21, SS06a, Hoc69]

2. BACKGROUND

[Sim06a, Sex03, Joh82]

3. COMPACT QUOTIENTS

Let $f: A \rightarrow B$ be a frame morphism with right adjoint $f_*: B \rightarrow A$. If $F \subseteq A$ and $G \subseteq B$ are filters, we define:

$$(1) \quad b \in f^*F \iff f_*(b) \in F \quad a \in f_*G \iff f(a) \in G,$$

where f^*F and f_*G are filters in B and A , respectively.

Proposition 3.1. *For $f = f^*: A \rightarrow B$ a frame morphism and $G \in B^\wedge$, then $f_*G \in A^\wedge$.*

Proof. By (1), f_*G is a filter in A . We need that f_*G satisfy the open filter condition. Let $X \subseteq A$ be such that $\bigvee X \in f_*G$, with X directed. Then

$$Y = \{f(x) \mid x \in X\}$$

is directed and $f(\bigvee X) = \bigvee f[X] = \bigvee Y \in G$. Since G is an open filter and $f(\bigvee X) = \bigvee f[X] \in G$, there exists $x \in X$ such that $f(x) \in G$. Therefore, $x \in f_*G$. $\square$

In [Sim04, Lemma 8.9 and Corollary 8.10] show that the diagram

$$(2) \quad \begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ U_A \downarrow & & \downarrow U_A \\ \mathcal{O}S & \xrightarrow{F^\infty} & \mathcal{O}S \end{array}$$

commutes laxly, that is, $U_A \circ f^\infty \leq F^\infty \circ U_A$. In this diagram f^∞ and F^∞ represent the fitted nuclei associated with the filters $F \in A^\wedge$ and $\nabla \in \mathcal{O}S^\wedge$, respectively, that is,

$$f^\infty = v_F \quad \text{and} \quad F^\infty = v_\nabla.$$

Here we prove something more general, since we consider $j \in NA$, $F \in A_j^\wedge$ and $j_*F \in A^\wedge$ to produce the diagrama

$$\begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ j^* \downarrow & & \downarrow j^* \\ A_j & \xrightarrow{\hat{f}^\infty} & A_j \end{array}$$

where $\hat{f}^\infty$ and f^∞ are the fitted nuclei associated with the filters j_*F and F , respectively.

The following lemma is part of the analysis developed in this work and constitutes a key tool in the construction of the commutative diagrams that follow.

Lemma 3.2. *For $j = j^*$, f and $\hat{f}$ as before, it holds that $j \circ \hat{f} \leq f \circ j$.*

Proof. Recall that, by (1)

$$\hat{f} = \bigvee \{v_y \mid y \in j_*F\} \quad f = \bigvee \{v_{j(y)} \mid j(y) \in F\}.$$

then for $a \in A$ it holds

$$v_y(a) = (y \succ a) \leq \hat{f}(a) \leq j(\hat{f}(a)).$$

Also, for all $a, y \in A$, $(y \succ a) \wedge y = y \wedge a$ and

$$\begin{aligned} j((y \succ a) \wedge y) \leq j(a) &\Leftrightarrow j(y \succ a) \wedge j(y) \leq j(a) \\ &\Leftrightarrow j(y \succ a) \leq (j(y) \succ j(a)). \end{aligned}$$

Thus,

$$v_y(a) \leq j(\hat{f}(a)) \leq (j(y) \succ j(a)) = v_{j(y)}(j(a)) \leq f(j(a)).$$

Therefore $j \circ \hat{f} \leq f \circ j$. $\square$

We now prove the above for an arbitrary ordinal.

Corollary 3.3. *For j , f and $\hat{f}$ as before, it holds that $j \circ \hat{f}^\alpha \leq f^\alpha \circ j$*

Proof. For $\alpha \in \text{Ord}$ we will verify that $j \circ \hat{f}^\alpha \leq f^\alpha \circ j$. We will do this with transfinite induction.

If $\alpha = 0$, it is trivial.

For the induction step, assume that for α the inequality holds. Then

$$j \circ \hat{f}^{\alpha+1} = j \circ \hat{f} \circ \hat{f}^\alpha \leq f \circ j \circ \hat{f}^\alpha \leq f \circ f^\alpha \circ j = f^{\alpha+1} \circ j,$$

where the first inequality is Lemma 3.2 and the second is the induction hypothesis.

If λ is a limit ordinal, then

$$\hat{f}^\lambda = \bigvee \{\hat{f}^\alpha \mid \alpha < \lambda\}, \quad f^\lambda = \bigvee \{f^\alpha \mid \alpha < \lambda\}$$

and

$$j \circ \hat{f}^\lambda = j \circ \bigvee_{\alpha < \lambda} \hat{f}^\alpha \leq \bigvee_{\alpha < \lambda} j \circ \hat{f}^\alpha.$$

By the induction hypothesis we have that

$$j \circ \hat{f}^\alpha \leq f^\alpha \circ j \Rightarrow \bigvee_{\alpha < \lambda} j \circ \hat{f}^\alpha \leq \bigvee_{\alpha < \lambda} f^\alpha \circ j.$$

Therefore $j \circ \hat{f}^\lambda \leq f^\lambda \circ j$. □

Observation 3.4. Lemma 3.2 and the Corollary 3.3 also hold if j is only monotone and preserves finite infima.

Remembering that $v_F = f^\infty$ and $v_{j_*F} = \hat{f}^\infty$, the Corollary 3.3 tells us that

$$j \circ v_{j_*F} \leq v_F \circ j$$

The objective now is to compare the compact quotients induced by F and by j_*F , and to construct a morphism between them that makes the corresponding diagram commutative.

If we consider j, v_F and v_{j_*F} we have the following diagram

$$(3) \quad \begin{array}{ccc} A & \begin{array}{c} \xrightarrow{(v_{j_*F})^*} \\ \xleftarrow{(v_{j_*F})_*} \end{array} & A_{j_*F} \\ \downarrow j & \searrow H & \\ A_j & \begin{array}{c} \xleftarrow{(v_F)_*} \\ \xrightarrow{(v_F)^*} \end{array} & A_F \end{array}$$

where A_F and A_{j_*F} are the compact quotients produced by v_F and v_{j_*F} , respectively.

We take the morphism H as the composition of v_F and j , that is,

$$H: A \rightarrow A_F, \quad H = v_F \circ j.$$

We want to define a morphism of frames between the compact quotients A_{j_*F} and A_F such that (3) be a commutative square.

Define the morphism h as the restriction of H to A_{j_*F} , that is,

$$h: A_{j_*F} \rightarrow A_j, \quad h = H|_{A_{j_*F}},$$

that is, $h(x) = H(x)$ for all $x \in A_{j_*F}$.

Note that, by Corollary 3.3, it holds that $H(A_{j_*F}) \subseteq A_F$. Thus, if $h = H|_{A_{j_*F}}$, the diagram

$$\begin{array}{ccc} A & \xrightarrow{v_{j_*F}} & A_{j_*F} \\ j \downarrow & & \downarrow h \\ A_j & \xrightarrow{v_F} & A_F \end{array}$$

is commutative.

We must verify that h is a frame morphism. By the definition of h , it is a $\wedge$ -morphism. It would remain to check that it is a $\bigvee$ -morphism.

The suprema in A_{j_*F} and A_F are calculated in a different way, namely

$$\bigvee_{v_{j_*F}} X = v_{j_*F}(\bigvee X) \quad \text{and} \quad \bigvee_{v_F} Y = v_F(\bigvee_j Y)$$

where $X \subseteq A$, $Y \subseteq A_j$.

Since H is a frame morphism, $H \circ \bigvee = \bigvee_{v_F} \circ H$. We need something similar for the morphism h .

Lemma 3.5. *Following the above information, it holds that*

$$h \circ \bigvee_{v_{j_*F}} = \bigvee_{v_F} \circ h.$$

Proof. It suffices to prove that

$$h \circ \bigvee_{v_{j_*F}} \leq \bigvee_{v_F} \circ h$$

since the reverse inequality follows from the definition of suprema in quotient frames.

First

$$h \circ \bigvee_{v_{j_*F}} = H \circ v_{j_*F} \circ \bigvee = v_F \circ j \circ v_{j_*F} \circ \bigvee \leq v_F \circ v_F \circ j \circ \bigvee$$

where the inequality is the Corollary 3.3. Furthermore, by the idempotency of v_F

$$h \circ \bigvee_{v_{j_*F}} \leq v_F \circ j \circ \bigvee = H \circ \bigvee = \bigvee_{v_F} \circ H = \bigvee_{v_F} \circ h.$$

Therefore $h \circ \bigvee_{v_{j_*F}} = \bigvee_{v_F} \circ h$. $\square$

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

Proposition 3.6. *The diagram*

$$(4) \quad \begin{array}{ccc} A & \xrightarrow{v_{j_*F}} & A_{j_*F} \\ j \downarrow & & \downarrow h \\ A_j & \xrightarrow{v_F} & A_F \end{array}$$

is commutative.

A particular case of the diagram (4) can be obtained by considering $j^* = \mathcal{U}_A$, that is,

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ \mathcal{U}_A \downarrow & & \downarrow h \\ \mathcal{O}S & \xrightarrow{v_\nabla} & \mathcal{O}S_\nabla \end{array}$$

It can be proven that $\text{pt } A_F = Q$, thus we have the spatial reflection

$$\mathcal{U}_{A_F} : A_F \rightarrow \mathcal{O} \text{pt } A_F = \mathcal{O}Q$$

and this is a quotient of A through v_F .

Also, we have the frame morphism

$$A \xrightarrow{\mathcal{U}_A} \mathcal{O}S \xrightarrow{v_\nabla} \mathcal{O}S_\nabla \xrightarrow{\mathcal{U}_{\mathcal{O}S_\nabla}} \mathcal{O}Q$$

and this is a quotient of A through v_∇ , since $\text{pt } \mathcal{O}S_\nabla = Q$. Since $Q \subseteq S$, both quotients are given by the assignment

$$x \mapsto \mathcal{U}_A(x) \cap Q$$

for $x \in A$.

The construction culminates in the following diagram, which synthesizes the interaction between compact quotients, separation, and lifts of nuclei.

$$(5) \quad \begin{array}{ccccc} A & \xrightarrow{v_F} & A_F & & \\ \mathcal{U}_A \downarrow & & \downarrow h & \searrow \mathcal{U}_{A_F} & \\ & & & & \mathcal{O}Q \\ \mathcal{O}S & \xrightarrow{v_\nabla} & \mathcal{O}S_\nabla & \nearrow \mathcal{U}_{\mathcal{O}S_\nabla} & \end{array}$$

Since the above diagram is codified through $F \in A^\wedge$ and the associated $Q \in \mathcal{Q}S$, we denote (5) as the Q -square.

Theorem 3.7. *Under the hypotheses of the Q -square (5), if $A \in \text{Frm}$ satisfies property **(H)**, then*

$$\mathcal{O}Q \cong \uparrow Q'.$$

Equivalently, the frame of open sets of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. By hypothesis, A satisfies **(H)**. Since **(H)** property is closed under quotient, A_F and $\mathcal{O}S$ satisfy **(H)**. Furthermore, **(H)** is conservative, that is, $S = \text{pt } A$ is an T_2 space. To be $S T_2$, if $Q \in \mathcal{Q}S$, then $Q \in \mathcal{C}S$. Thus,

$$S \text{ is } T_2 \Rightarrow S \text{ is strongly stacked} \Rightarrow v_\nabla = [Q'].$$

Because $Q' \in \mathcal{O}S$, $[Q'] = u_{Q'}$ and this is complemented in $N\mathcal{O}S$. Additionally,

$$\mathcal{O}S_{u_{Q'}} = \uparrow Q' = \mathcal{O}S_{v_\nabla}.$$

By [PP12, Proposition VI.3.3], we have that $\mathcal{O}S_{v_\nabla}$ is spatial. From this, h is an isomorphism.

Therefore, $\mathcal{O} \text{pt } A_F \cong \mathcal{O}S_\nabla$, that is, $\mathcal{O}Q \cong \uparrow Q'$. □

4. TIDY FRAMES

Proposition 4.1. *If A is a tidy frame, then A_j is a tidy frame.*

Proof. Let us first observe that $F \subseteq j_*F$. Since A is tidy and $F \in A^\wedge$, it holds that

$$x \in F \Rightarrow \hat{d} \vee x = 1,$$

where $\hat{d} = d(\alpha) = f^\alpha(0)$.

If $\hat{d} \leq d$, then $d \vee x = 1$, for $d = d(\alpha) = f^\alpha(j(0))$.

Thus, by the Corollary 3.3

$$\hat{d} = \hat{d}(\alpha) \leq j(\hat{d}(\alpha)) = j(\hat{f}^\alpha(0)) \leq f^\alpha(j(0)) = d(\alpha) = d.$$

Therefore, if $x \in F$, then $d \vee x = 1$ and A_j is tidy. □

The following theorem establishes that tidiness is preserved under binary coproducts.

Proposition 4.2. *If $A_1, A_2 \in \text{Frm}$ are tidy frames, then $A_1 \oplus A_2$ is a tidy frame.*

Proof. Let A_1, A_2 be α -tidy frames. By construction of the coproduct $A_1 \oplus A_2$, we have the following diagram

$$\begin{array}{ccccccc}
 & & \xrightarrow{\quad \iota_i \quad} & & & & \\
 A_i & \xrightarrow{\quad \alpha_i \quad} & A_1 \times A_2 & \xrightarrow{\quad \downarrow(-) \quad} & \mathcal{L}(A_1 \times A_2) & \xrightarrow{\quad \sim \quad} & A_1 \oplus A_2 \\
 & & \xleftarrow{\quad (\iota_i)_* \quad} & & & &
 \end{array}$$

where $\alpha_1(a) = (a, 1)$ and $\alpha_2(b) = (1, b)$ to $a \in A_1$ and $b \in A_2$. Furthermore, as $\iota_i \in \text{Frm}$, $(\iota_i)_*$ is its right adjoint.

Take $F \in (A_1 \oplus A_2)^\wedge$ and, by Proposition 3.1, $(\iota_i)_*[F] \in A_i^\wedge$. As A_i is tidy, for $x_i \in (\iota_i)_*F$ it holds that $d_i \vee x_i = 1$, where

$$d_i = d_i(\alpha) = f_i^\alpha(0), \text{ con } f_i = \bigvee \{v_y \mid y \in (\iota_i)_*F\}$$

By (1) we have that if $x_i \in (\iota_i)_*F$, then $\iota_i(x_i) \in F$, that is,

$$x_1 \oplus 1, 1 \oplus x_2 \in F$$

and because F is a filter, we have that $x_1 \oplus 1 \wedge 1 \oplus x_2 = x_1 \oplus x_2 \in F$.

Additionally, for each $i = 1, 2$ we have the following diagram

$$\begin{array}{ccc}
 A_i & \xrightarrow{\quad f_i \quad} & A_i \\
 \iota_i \downarrow & & \downarrow \iota_i \\
 A_1 \oplus A_2 & \xrightarrow{\quad f_F \quad} & A_1 \oplus A_2
 \end{array}$$

where $f_F = \bigvee \{v_{\iota_i(y)} \mid \iota_i(y) \in F\}$.

Let $a \in A_i$ be, then $v_y(a) \leq f_i(a)$ and $\iota_i(v_y(a)) \leq \iota_i(f_i(a))$. Additionally,

$$\begin{aligned}
 \iota_i((y \succ a) \wedge y) &= \iota_i(y \wedge a) \leq \iota_i(a) \Leftrightarrow \iota_i(y \succ a) \wedge \iota_i(y) \leq \iota_i(a) \\
 &\Leftrightarrow \iota_i(y \succ a) \leq (\iota_i(y) \succ \iota_i(a)).
 \end{aligned}$$

Next

$$\iota_i(f_i(a)) \leq (\iota_i(y) \succ \iota_i(a)) = v_{\iota_i(y)}(\iota_i(a)) \leq f_F(\iota_i(a)).$$

Therefore, $\iota_i \circ f_i \leq f_F \circ \iota_i$.

We prove by transfinite induction that $\iota_i \circ f_i^\alpha \leq f_F^\alpha \circ \iota_i$ with $\alpha \in \text{Ord}$.

- If $\alpha = 0$, it is trivial.
- For the induction step, assume that for α the inequality holds. Then

$$\iota_i \circ f_i^{\alpha+1} = \iota_i \circ f_i \circ f_i^\alpha \leq f_F \circ \iota_i \circ f_i^\alpha \leq f_F \circ f_F^\alpha \circ \iota_i = f_F^{\alpha+1} \circ \iota_i.$$

- If λ is a limit ordinal, then

$$f_i^\lambda = \bigvee \{f_i^\alpha \mid \alpha < \lambda\}, \quad f_F^\lambda = \bigvee \{f_F^\alpha \mid \alpha < \lambda\}$$

and

$$\iota_i \circ f_i^\lambda = \iota_i \circ \bigvee_{\alpha < \lambda} f_i^\alpha \leq \bigvee_{\alpha < \lambda} \iota_i \circ f_i^\alpha.$$

By the induction hypothesis we have that

$$\iota_i \circ f_i^\alpha \leq f_F^\alpha \circ \iota_i \Rightarrow \bigvee_{\alpha < \lambda} \iota_i \circ f_i^\alpha \leq \bigvee_{\alpha < \lambda} f_F^\alpha \circ \iota_i.$$

Evaluating at $0 \in A_i$, we obtain that

$$(\iota_i \circ f_i^\alpha)(0) \leq (f_F^\alpha \circ \iota_i)(0) \Rightarrow \iota_1(d_1) \leq f_F^\alpha(0 \oplus 1) \text{ y } \iota_2(d_2) \leq f_F^\alpha(1 \oplus 0).$$

Thus

$$d_1 \oplus d_2 = d_1 \oplus 1 \wedge 1 \oplus d_2 \leq f_F^\alpha(0 \oplus 1) \wedge f_F^\alpha(1 \oplus 0) = f_F^\alpha(0 \oplus 0).$$

For $d = d(\alpha) = f_F^\alpha(0 \oplus 0)$,

$$(x_1 \oplus x_2) \vee d_1 \oplus d_2 \leq (x_1 \oplus x_2) \vee d,$$

but $x_1 \oplus x_2 \vee d_1 \oplus d_2 = (x_1 \vee d_1) \oplus (x_2 \vee d_2) = 1 \oplus 1$.

Therefore,

$$x_1 \oplus x_2 \vee d = 1 \oplus 1,$$

that is, $A_1 \oplus A_2$ is tidy. $\square$

Corollary 4.3. *If $A_1, A_2 \in \text{Frm}$ are α -tidy and β -tidy, respectively, then $A_1 \oplus A_2$ is tidy.*

Proof. Let $\gamma = \max\{\alpha, \beta\}$ be, then A_1 and A_2 are γ -tidy and by Proposition 4.2, $A_1 \oplus A_2$ is tidy. $\square$

We extend the previous result to arbitrary coproducts

Proposition 4.4. *Tidiness is preserved under coproducts.*

Proof. Let $\{A_i\}_{i \in I}$ be a family of tidy frames and $\{\alpha_i\}_{i \in I} \subseteq \text{Ord}$ the efficiency degrees of each A_i . We know that

$$\bigoplus_{i \in I} A_i \in \text{Frm} \quad \text{and} \quad (\iota_i : A_i \rightarrow \bigoplus_{i \in I} A_i) \in \text{Frm},$$

then $(\iota_i)_*$ is the right adjoint of ι_i . Additionally, for $F \in (\bigoplus_{i \in I} A_i)^\wedge$, by Proposition 3.1, $(\iota_i)_* F \in A_i^\wedge$ and by (1),

$$x_i \in (\iota_i)_* F \Leftrightarrow \iota_i(x_i) \in F.$$

Take $\alpha = \sup\{\alpha_i \mid i \in I\}$, then each A_i is α -tidy. By hypothesis, for $x_i \in (\iota_i)_* F$ it holds that $x_i \vee d_i = 1$, where

$$d_i = d_i(\alpha) = f_i^\alpha(0), \text{ with } f_i = \bigvee \{v_y \mid y \in (\iota_i)_* F\}.$$

Claim: For all $x_i \in (\iota_i)_* F$, $\langle x_i \rangle = \bigoplus \iota_i(x_i) \in F$.

We know that $\bigoplus \iota_i(x_i) = \bigvee \iota_i(x_i)$. Then, $\iota_i(x_i) \leq \bigvee \iota_j(x_j)$ for all $i \in I$ and $j \in I$. Therefore, because F is a filter, we have that $\bigvee \iota_i(x_i) \in F$.

For each A_i there exists the following diagram

$$\begin{array}{ccc} A_i & \xrightarrow{f_i} & A_i \\ \downarrow \iota_i & & \downarrow \iota_i \\ \bigoplus_{i \in I} A_i & \xrightarrow{f_F} & \bigoplus_{i \in I} A_i \end{array}$$

where $f_F = \bigvee \{v_{\iota_i(y)} \mid \iota_i(y) \in F\}$.

Proceeding as in the proof of Proposition 4.2, we obtain

$$\iota_i \circ f_i \leq f_F \circ \iota_i$$

and, by transfinite induction,

$$\iota_i \circ f_i^\alpha \leq f_F^\alpha \circ \iota_i.$$

Evaluating at $0 \in A_i$, we obtain

$$(\iota_i \circ f_i^\alpha)(0) \leq (f_F^\alpha \circ \iota_i)(0) \iff \iota_i(d_i) \leq f_F^\alpha(\langle 0 \rangle) = d(\alpha) = d.$$

and $\langle d_i \rangle = \bigoplus \iota_i(d_i) \leq d$.

Thus, $\langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d$, but

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle,$$

that is, $\langle 1 \rangle \leq \langle x_i \rangle \vee d$ and therefore $\langle x_i \rangle \vee d = \langle 1 \rangle$. By above claim, $\bigoplus_{i \in I} A_i$ is tidy. $\square$

Corollary 4.5. *The subcategory of tidy frames is a coreflective subcategory of $\mathbf{Frm}$.*

5. STACKED AND STRONGLY STACKED FRAMES

Lemma 5.1. *If $\alpha \in \mathbf{Ord}$ and $X \in CS$, then*

$$(6) \quad \partial_F^\alpha(X) = (f_F^\alpha(X'))'.$$

Proof. By transfinite induction on $\alpha \in \mathbf{Ord}$:

- If $\alpha = 0$, then $\partial_F^0 = \text{id}$, that is, $\partial_F^0(X) = X$ and $f_F^0 = \text{id}$. Thus

$$\partial_F^0(X) = X = (X'') = (f_F^0(X'))'.$$

- Assume that $\partial_F^\alpha(X) = (f_F^\alpha(X'))'$ holds. By definition,

$$\partial_F^{\alpha+1}(X) = \partial_F(\partial_F^\alpha(X)).$$

Furthermore,

$$\partial_F^{\alpha+1}(X) = (f_F((\partial_F^\alpha(X))'))'.$$

By induction hypothesis, $\partial_F^\alpha(X) = (f_F^\alpha(X'))'$ and substituting we obtain

$$\partial_F^{\alpha+1}(X) = (f_F(f_F^\alpha(X')))' = (f_F^{\alpha+1}(X'))'$$

which is what we wanted.

- Let λ be a limit ordinal and assume that the statement holds for all $\alpha < \lambda$.
By definition,

$$\partial_F^\lambda(X) = \bigcap_{\alpha < \lambda} \partial_F^\alpha(X).$$

By De Morgan laws, we have that

$$(\partial_F^\lambda(X))' = \left(\bigcap_{\alpha < \lambda} \partial_F^\alpha(X) \right)' = \bigcup_{\alpha < \lambda} (\partial_F^\alpha(X))'.$$

Using the induction hypothesis, for all $\alpha < \lambda$,

$$(\partial_F^\alpha(X))' = f_F^\alpha(X'),$$

Then,

$$(\partial_F^\lambda(X))' = \bigcup_{\alpha < \lambda} f_F^\alpha(X') = f_F^\lambda(X').$$

Therefore,

$$(\partial_F^\lambda(X))' = f_F^\lambda(X'),$$

that is,

$$\partial_F^\lambda(X) = (f_F^\lambda(X'))'.$$

□

As a particular case, we have the following.

Corollary 5.2. *If $X \in \mathcal{CS}$, $U \in \mathcal{OS}$ and $\alpha = \infty$, then*

$$(7) \quad \begin{aligned} \partial_F^\infty(X) = (v_F(X'))' &\iff \partial_F^\infty(U')' = (v_F(U)) \\ &\iff \partial_F^\infty(X)' = (v_F(X')). \end{aligned}$$

Lemma 5.3. *If $A = \mathcal{OS}$ and $E \subseteq S$, then*

$$\mathcal{U}_*[E]\mathcal{U} = [E].$$

Proof. Since A is spacial, then $\mathcal{U}$ is an isomorphism. Thus, if $U \in \mathcal{O} \text{ pt } A$ exists $V \in \text{pt } A$ such that $U = \mathcal{U}(V)$ and as $U \in \text{pt } A$, then $U = V$. Next,

$$\begin{aligned} (\mathcal{U}_*[E]\mathcal{U})(U) &= \mathcal{U}_*[E](\mathcal{U}(U)) = \mathcal{U}_*([E](U)) \\ &= \mathcal{U}_*(E \cup U)^\circ = \bigcup \{W \in \mathcal{OS} \mid W \subseteq (E \cup U)^\circ\} \\ &= (E \cup U)^\circ = [E](U). \end{aligned}$$

Therefore $\mathcal{U}_*[E]\mathcal{U} = [E]$. □

Proposition 5.4. *For $F \in A^\wedge$, $Q \in \mathcal{QS}$ and $\nabla = \nabla(Q)$, if $k \in [v_\nabla, w_\nabla]$, then $\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F]$.*

Proof. If $k \in [v_\nabla, w_\nabla]$, then $\mathcal{U}_* \circ k \circ \mathcal{U} \in NA$. Additionally, for $F \in A^\wedge$ if $j \in [v_F, w_F]$, then $F = \nabla(j)$. In this way we must prove that $F = \nabla(\mathcal{U}_* \circ k \circ \mathcal{U})$.

By Hofmann-Mislove Theorem, we have that

$$x \in F \iff Q \subseteq \mathcal{U}(x) \iff \mathcal{U}(x) \in \nabla = \nabla(Q).$$

Furthermore, for all $x \in A$

$$(\mathcal{U}_* \circ k \circ \mathcal{U})(x) = \mathcal{U}_*(k(\mathcal{U}(x))) = \bigwedge (S \setminus k(\mathcal{U}(x))).$$

Since $k \in [v_\nabla, w_\nabla]$, then $\nabla = \nabla(k)$, that is, for $V \in \nabla$ it holds that $k(V) = S$. Therefore,

$$\begin{aligned} x \in F &\iff \mathcal{U}(x) \in \nabla \iff S \setminus k(\mathcal{U}(x)) = \emptyset \\ &\iff \bigwedge (S \setminus k(\mathcal{U}(x))) = 1 \\ &\iff x \in \nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) \end{aligned}$$

□

We have a morphism $\mathcal{U}: [v_\nabla, w_\nabla] \rightarrow [v_F, w_F]$. Note that this occurs at both ends of the interval on NA .

Take $k \in [v_\nabla, w_\nabla]$. Since v_F is a fitted nuclei, then it holds

$$(8) \quad v_F \leq \mathcal{U}_* \circ k \circ \mathcal{U}.$$

Similarly, as w_F is the largest element in its block,

$$(9) \quad \mathcal{U}_* \circ k \circ \mathcal{U} \leq w_F.$$

Proposition 5.5. *If $F \in A^\wedge$ and $\nabla \in \mathcal{OS}^\wedge$, then*

$$w_F = \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}.$$

Proof. By (9), we must verify that $w_F \leq \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}$. Equivalently, we must prove that

$$\mathcal{U} \circ w_F \leq w_\nabla \circ \mathcal{U},$$

that is, for all $x \in A$

$$\mathcal{U}(w_F(x)) \subseteq [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Since $\mathcal{U}(w_F(x)) \in \mathcal{OS}$, it is enough that $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$. As a contradiction, let $y \in \mathcal{U}(w_F(x))$ be such that $y \notin M'$ and $y \notin \mathcal{U}(x)$. Then

$$y \in M \quad \text{and} \quad x \leq y.$$

By hypothesis, $y \in \mathcal{U}(w_F(x))$, then

$$w_F(x) \not\leq y \quad \text{and} \quad w_F(x) = \bigwedge \{m \in M \mid x \leq m\},$$

that is, $w_F(x) \leq y$ and it is a contradiction.

Therefore, $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$.

□

Definition 5.6. Let A be a frame.

- (1) A is *strongly stacked* if $\mathcal{U}_*[Q']\mathcal{U} = v_F$.
- (2) A is *stacked* if $(\mathcal{U}_*([Q'])\mathcal{U})(0) = v_F(0)$.

for any (F, Q) HM pair.

Note that strongly stacked implies stacked. Our aim is to prove that the above properties are conservative, that is, S is a stacked (strongly stacked) space if and only if its frame of open sets $\mathcal{O}S$ is stacked (strongly stacked).

Proposition 5.7. *The notions of strongly stacked and stacked are conservative.*

Proof. Consider $S \in \mathbf{Top}$, $A = \mathcal{O}S$, $F \in A^\wedge$ and $Q \in \mathcal{Q}S$.

First, suppose that S is strongly stacked, then $v_F = [Q']$ and, by Lemma 5.3, $\mathcal{U}_*[Q']\mathcal{U} = [Q'] = v_F$. Therefore $\mathcal{O}S$ is strongly stacked.

Now, suppose that S is stacked, then

$$\overline{Q} = \partial_F^\infty(S) \iff \overline{Q}' = (\partial_F^\infty(S))'.$$

Furthermore,

$$\begin{aligned} (\mathcal{U}_*[Q']\mathcal{U})(\emptyset) &= \mathcal{U}_*([Q'](\{V \in \mathbf{pt} \mathcal{O}S \mid \emptyset \not\subseteq V\})) \\ &= \mathcal{U}_*([Q'](\emptyset)) = \mathcal{U}_*(Q')^\circ \\ &= \bigcup \{W \in \mathcal{O}S \mid W \subseteq (Q')^\circ\} \\ &= (Q')^\circ = \overline{Q}'. \end{aligned}$$

By Corollary 5.2, $v_F(\emptyset) = (\partial_F^\infty(\emptyset'))'$. Therefore,

$$v_F(\emptyset) = (\mathcal{U}_*[Q']\mathcal{U})(\emptyset),$$

and $\mathcal{O}S$ is stacked. □

6. SEPARATION PROPERTIES IN FRAMES

If $S \in \mathbf{Top}$ is an T_1 and strongly stacked, then it holds that $w_\nabla = v_\nabla$. Additionally, if S is stacked and T_1 , then it is sober.

Thus, we could ask ourselves if the same occurs in frames. In this section we will study the above question.

Take $p \in \mathbf{pt} A$ and $(F_p, \uparrow p)$ a HM pair, then for

$$M = \{m \in A \setminus F_p \mid (\forall a \in A \setminus F_p)[a \leq m]\},$$

the nuclei $w_{F_p}(x) = \bigwedge \{m \in M \mid x \leq m\}$ is $w_p(x)$.

To prove this, note that $A \setminus F_p = \{x \in A \mid x \leq p\} = \downarrow p$. As,

$$M = \{m \in \downarrow p \mid [\forall a \in \downarrow p](a \leq m)\} = \{p\}.$$

Hence, $w_{F_p}(x) = \bigwedge \{p \mid x \leq p\}$, that is,

$$(10) \quad w_{F_p}(x) = \begin{cases} 1, & \text{if } x \not\leq p \\ p, & \text{if } x \leq p \end{cases} = w_p(x)$$

for all $x \in A$. Therefore $w_{F_p} = w_p$.

Proposition 6.1. *Let $A \in \text{Frm}$ and (F, Q) be any HM pair. If $v_F(0) = w_F(0)$, then for all $p \in \text{pt } A$, p is maximal in $\text{pt } A$.*

Proof. Consider $p, q \in \text{pt } A$ such that $p \leq q$. Note that for $q \in \text{pt } A$, we have $(F_q, \uparrow q)$ a HM pair, with

$$F_q = \{x \in A \mid \uparrow q \subseteq \mathcal{U}(x)\} = \{x \in A \mid (\forall r \in \text{pt } A)[r \leq q \Rightarrow x \not\leq r]\}.$$

By hypothesis, $v_{F_q}(0) = w_{F_q}(0)$ and by (10)

$$q = w_q(0) = w_{F_q}(0) = v_{F_q}(0) \leq v_{F_q}(p) = p.$$

Therefore, $p = q$ and every point is maximal in $\text{pt } A$. $\square$

As we show later on (2), for all (F, Q, ∇) HM tupla we have

$$\mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U}.$$

Additionally, if A is strongly stacked, then $v_F = \mathcal{U}_*[Q']\mathcal{U}$ and

$$\mathcal{U}_* \circ v_\nabla \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q']\mathcal{U}$$

Thus, $v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}$, equivalently $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$.

Therefore, if A is strongly stacked, then the diagram (2) is commutative. The following result tells us under which conditions we have the converse implication.

Proposition 6.2. *If $A \in \text{Frm}$ is tidy and the diagram*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ \mathcal{U} \downarrow & & \downarrow \mathcal{U} \\ \mathcal{OS} & \xrightarrow{v_\nabla} & \mathcal{OS} \end{array}$$

commutes, then A is strongly stacked.

Proof. Let $A \in \text{Frm}$, $S = \text{pt } A$ and (F, Q) be any HM pair. By hypothesis, A is tidy and by previous results we have the implications

$$A \text{ tidy} \Rightarrow \mathcal{OS} \text{ tidy} \Rightarrow S \text{ stacked and packed}.$$

Since S is packed, if $Q \in \mathcal{QS}$, then $Q \in \mathcal{CS}$ and $Q' \in \mathcal{OS}$. Then, as S is stacked, we have that $\overline{Q} = \partial_\nabla^\infty(S)$, that is, $Q' = v_\nabla(\emptyset)$ and as $\mathcal{OS}$ is tidy, $v_\nabla = u_{Q'} = [Q']$.

Additionally, if $v_\nabla \circ \mathcal{U} = \mathcal{U} \circ v_F$, then

$$v_\nabla \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{and} \quad \mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U},$$

that is,

$$[Q'] \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{and} \quad \mathcal{U} \circ v_F \leq [Q'] \circ \mathcal{U}.$$

By properties of the right adjoint

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq v_F \quad \text{and} \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$ and A is strongly stacked. $\square$

Consequently, we have the following result.

Corollary 6.3. *If A is tidy, then $S = \text{pt } A$ is strongly stacked.*

Proof. Since S is strongly stacked if and only if $v_\nabla = [Q']$, the result follows from the fact that

$$\mathcal{O}S \text{ tidy} \Rightarrow S \text{ strongly stacked}$$

$\square$

With the morphism $\mathcal{U}_A$, we have the nuclei sp_A defined by

$$\text{sp}_A(x) = \bigwedge \{p \in \text{pt } A \mid x \leq p\}$$

for $x \in A$. Combining this fact with the Proposition 6.1 we know information about the collapse on admisibility interval.

Proposition 6.4. *For $A \in \text{Frm}$ and (F, Q) any HM pair, the following statements hold:*

- (1) *If A is T_1 and stacked, then $v_F(0) = w_F(0)$.*
- (2) *If $v_F(0) = w_F(0)$, then A is stacked.*
- (3) *If $v_F(0) = w_F(0)$ and the set $Y = \{q \in \text{pt } A \mid a \leq q\}$ is non-empty for all $a \in A$, then A is T_1 .*

Proof.

- (1) If A is T_1 , then

$$\mathcal{O}S \text{ is } T_1 \Rightarrow S \text{ is } T_1 \Rightarrow Q = M.$$

Since A is stacked,

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) = w_F(0).$$

- (2) Recall that for every HM pair, the following inequality holds:

$$(11) \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F$$

and by hypothesis $v_F(0) = w_F(0)$, that is, $v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = w_F(0)$. Therefore, A is stacked.

- (3) Consider $p \in \text{pt } A$ and $a \in A$ such that $p \leq a$. By hypothesis $\{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$. Then

$$p \leq a \leq \text{sp}(a) = \bigwedge \{q \in \text{pt } A \mid a \leq q\}$$

for some $q \in Y$. Thus, $p \leq a \leq q$ and by Proposición 6.1, $p = q$. Therefore, $p = a$ and every point is maximal in A , that is, A is T_1 . $\square$

Corollary 6.5. *For $A \in \text{Frm}$ and (F, Q) any HM pair, the following statements hold:*

- (1) *If A is T_1 and strongly stacked, then $v_F = w_F$.*
- (2) *If $v_F = w_F$, then A is strongly stacked.*
- (3) *If $v_F = w_F$ and the set $Y = \{q \in \text{pt } A \mid a \leq q\}$ is non-empty for all $a \in A$, then A is T_1 .*

Proof.

- (1) If A is strongly stacked, then $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$ and by T_1 we have that $Q = M$. Therefore $v_F = w_F$.
- (2) This follows from (11).
- (3) This follows from Proposition 6.4.

□

Note that, by 6.1 and 6.4, if $v_F(0) = w_F(0)$, then A is stacked and every point is maximal in $\text{pt } A$. Particularly, for each $p \in \text{pt } A$ and $(F_p, \uparrow p)$ it holds $v_{F_p}(0) = p$. Furthermore,

$$v_{F_p}(p) = v_{F_p}(0) = 0_{A_{F_p}}.$$

If A is T_1 , p is maximal in A , then p is maximal in A_{F_p} and $A_{F_p} = \{q, 1\}$.

Now, $F_{F_p} = \nabla(v_{F_p})$ and $x \in F_p$ if and only if $v_{F_p}(x) = 1$. In this way, under the previous hypotheses, it holds that

$$v_{F_p}(x) = \begin{cases} 1, & \text{si } x \not\leq p \\ p, & \text{si } x \leq p \end{cases} = \begin{cases} 1, & \text{si } x \in F_p \\ p, & \text{si } x \notin F_p \end{cases} = w_{F_p}(x).$$

Therefore, if A is T_1 and stacked, then $v_{F_p} = w_{F_p}$ for each $p \in \text{pt } A$. With this in mind, we could ask ourselves if there exists some way to recover the notion of strongly stacked by means of T_1 and stacked.

Now, we need the following information:

- (1) Let $A^\wedge$ be a topological space where the basic open sets of $\mathcal{O}(A^\wedge)$ are given by

$$D(x) = \{F \in A^\wedge \mid x \in F\}$$

where $x \in A$.

- (2) For $A \in \text{Frm}$, we can construct a frame VA as in [Sim04]. Each element in $\text{pt}(VA)$ is of the form

$$(F, a), \quad \text{where } F \in A^\wedge \text{ y } a \in [v_F(0), w_F(0)].$$

- (3) Take a continuous surjective map $\pi: \text{pt}(VA) \rightarrow A^\wedge$, defined by

$$(F, a) \mapsto F.$$

- (4) By the functor $\mathcal{O}(-)$ we have

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \rightarrow \mathcal{O}(\text{pt}(VA)).$$

(5) Open sets in $\text{pt}(VA)$ are given by the following:

$$(12) \quad (F, a) \in [-](x) \iff x \in F \quad \text{and} \quad (F, a) \in \langle - \rangle(x) \iff x \not\leq a.$$

Particularly,

$$(13) \quad a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F \rangle(x) \iff x \not\leq a.$$

Proposition 6.6. *Let $A, VA \in \text{Frm}$ and $A^\wedge$ be as above. The following statements hold:*

- (1) *If A is T_1 and stacked, then $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ under the continuous function π .*
- (2) *If the morphism $\mathcal{O}(\pi) = \pi^{-1}$ induces that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$, then A is stacked.*
- (3) *If the morphism $\mathcal{O}(\pi) = \pi^{-1}$ induces that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ and*

$$\mathcal{V}(a) = \{q \in \text{pt } A \mid a \leq q\} \neq \emptyset,$$

then A is T_1 .

Proof.

- (1) If A is T_1 and stacked, then for each $F \in A^\wedge$ it holds that

$$v_F(0) = w_F(0).$$

Take $(F, a), (G, b) \in \text{pt}(VA)$ such that $\pi(F, a) = F = \pi(G, a) = G$. By T_1 and stacked we have

$$a = v_F(0) = v_G(0) = b.$$

Therefore $(F, a) = (G, b)$, that is, π is injective. Since it is also surjective, π is bijective.

We must show that $\pi^{-1}: A^\wedge \rightarrow \text{pt}(VA)$ is continuous. For this, recall that by (12)

$$\begin{aligned} (F, a) \in [-](x) &\iff x \in F \\ &\iff (F, a) \in \pi^{-1}(D(x)) \\ &\iff \pi([-](x)) = D(x). \end{aligned}$$

For the other basic open sets, by (13) we have

$$a \notin \langle F \rangle(x) \iff x \leq a \iff v_F(x) \leq a.$$

By T_1 and stacked we have that $a = v_F(0)$ and thus

$$v_F(0) \notin \langle F \rangle(x) \iff x \leq v_F(0) \iff v_F(x) = v_F(0).$$

As $v_F(0) \neq 1$,

$$v_F(0) \notin \langle F \rangle(x) \iff x \notin F$$

or equivalently

$$a \in \langle F \rangle(x) \iff (F, a) \in \langle - \rangle(x) \iff x \in F.$$

Therefore, $\pi(\langle - \rangle(x)) = D(x)$ y π^{-1} is continuous. Hence, π is a homeomorphism and

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA)).$$

- (2) Suppose that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ under $\mathcal{O}(\pi) = \pi^{-1}$. Since π^{-1} is bijective, then π is also bijective. Thus, if $F \in A^\wedge$, then

$$(F, v_F(0)), (F, w_F(0)) \in \text{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

By the injectivity of π we have that $v_F(0) = w_F(0)$ and therefore A is stacked.

- (3) Suppose that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ and A is stacked. Consider $a \in A$ and $p \in \text{pt } A$ such that $p \leq a$. Moreover, by the previous item we have that

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA)) \Rightarrow v_F(0) = w_F(0)$$

and every point is maximal in $\text{pt } A$. Since $\mathcal{V}(a) \neq \emptyset$, there exists $q \in \text{pt } A$ such that $a \leq q$. By the maximality of p and q , we have

$$p \leq a \leq q = p.$$

Therefore, $p = a$ and A is T_1 .

□

Note that item (3) holds whenever A is spatial. In the spatial setting, this corresponds to the topological fact that

$$S \text{ is } T_1 \text{ and stacked} \Rightarrow S \text{ is sober.}$$

However, the point-free setting allows for a potentially broader phenomenon, since the conditions of being T_1 and stacked do not a priori force a frame to be spatial. This leads naturally to the following question:

Do there exist T_1 stacked frames that are not spatial?

7. QUOTIENTS OF STACKED FRAMES

Remember that for $(f: A \rightarrow B) \in \text{Frm}$, $j \in NA$ and $k \in NB$, we have the following definitions:

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NA_j$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where f_* is the right adjoint of f .

Proposition 7.1. *Let A be a frame and $j \in NA$. If A is strongly stacked, then A_j is strongly stacked.*

Proof. Consider $S = \text{pt } A$, $S_j = \text{pt } A_j$ and let $F \in A_j^\wedge$ be. We define:

$$G = j_*(F), \quad Q = S_j \setminus F, \quad R = S \setminus G,$$

where j_* is the right adjoint associated with the frame morphism $j^*: A \rightarrow A_j$ and $x \in G$ if and only if $j(x) \in F$. Considering the spatial reflections

$$\mathcal{U}: A \rightarrow \mathcal{O}S \quad \text{and} \quad \mathcal{U}_j: A_j \rightarrow \mathcal{O}S_j$$

we have the following diagram

$$\begin{array}{ccccc}
 & & A & \xrightarrow{\mathcal{U}} & \mathcal{O}S \\
 & \nearrow v_G & \downarrow j^* & \nearrow [R'] & \downarrow sj \\
 A & \xrightarrow{\mathcal{U}} & \mathcal{O}S & & \\
 \downarrow j^* & & \downarrow sj & \nearrow v_F & \\
 A_j & \xrightarrow{\mathcal{U}_j} & \mathcal{O}S_j & \xrightarrow{\mathcal{U}_j} & \mathcal{O}S_j \\
 & \nearrow [Q'] & & &
 \end{array}$$

Using the right adjoints of $\mathcal{U}$ and $\mathcal{U}_j$ we obtain

$$v_F, \quad l = N(\mathcal{U}_j)_*([Q']), \quad m = N(j^*)(N(\mathcal{U})_*([R'])), \quad n = N(j^*)(N(\mathcal{U}_j \circ j^*)_*)([Q'])$$

are nuclei in NA_j . Let us see what is the relationship between these nuclei.

By construction, we have $v_F \leq N(\mathcal{U}_j)_*([Q'])$ and we must verify the other inequality for A_j to be strongly stacked.

By hypothesis, we have $v_G = N(\mathcal{U})_*([R'])$. Moreover, for $x \in A$, we have $j(v_x) \leq v_{j(x)}$ and, by definition of the adjusted nuclei associated with F and G ,

$$(14) \quad j \circ v_G \leq v_F.$$

Furthermore, for $(j: A \rightarrow A_j) \in \text{Frm}$

$$N(j^*)(v_G) = \bigvee \{u_{j(v_G(a))} \wedge v_{j(a)} \mid a \in A\} \quad \text{y} \quad v_F = \bigvee \{u_{v_F(a)} \wedge v_a \mid a \in A_j\}.$$

Note, by (14), $u_{j(v_G)} \leq u_{v_F}$ and therefore

$$(15) \quad m \leq v_F.$$

To l and n , we have

$$n = \bigvee \{u_{j^*(n(x))} \wedge v_{j^*(x)} \mid x \in A\} \quad \text{and} \quad l = \bigvee \{u_{l(x)} \wedge v_x \mid x \in A_j\},$$

where

$$j^*(n(x)) = j^*((j_* \circ \mathcal{U}_{j*} \circ [Q'] \circ \mathcal{U}_j \circ j^*)(x)) \quad \text{y} \quad l(x) = N(\mathcal{U}_j)_*([Q'])(x).$$

Then,

$$\begin{aligned} j^*(n(x)) &= j^*(j_*(\bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\})) \\ &= \bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}. \end{aligned}$$

Since $x \in A_j$,

$$l(x) = \bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}.$$

Therefore $j(n(x)) = l(x)$.

To $m(x)$ we have

$$m(x) = \bigvee\{u_{j^*}(\mathcal{U}_* \circ [R'] \circ \mathcal{U})(x) \wedge v_{j^*}(x) \mid x \in A\}.$$

Since $A_j \subseteq A$, then $Q = R \cap A_j$ and

$$m(x) = \bigvee\{u_{j^*}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x) \wedge v_x \mid x \in A_j\}.$$

Note

$$\begin{aligned} j^*(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x) &= j^*(\bigvee\{a \in A \mid \mathcal{U}(a) \subseteq Q' \cup \mathcal{U}(x)\}) \\ &= \bigvee\{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\}. \end{aligned}$$

To $x \in A_j$

$$\begin{aligned} \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(j^*(x)) &\iff \mathcal{U}(j^*(a)) \cap S_j \subseteq (Q' \cup \mathcal{U}(j^*(x))) \cap S_j \\ &\iff \mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(j^*(x)), \end{aligned}$$

that is,

$$\bigvee\{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\} = j^*(N(\mathcal{U}_j \circ j)_*)([Q']).$$

Thus, we have

$$\bigvee\{u_{j^*}(N(\mathcal{U}_j \circ j)_*)([Q'])(x) \wedge x \mid x \in A\} = m(x).$$

Since $(N(\mathcal{U}_j \circ j)_*)([Q']) \leq j(N(\mathcal{U}_j \circ j)_*)([Q'])$ and by (15)

$$l \leq m \leq v_F \leq l.$$

Therefore $v_F = \mathcal{U}_{j^*} \circ [Q'] \circ \mathcal{U}_j$, that is, A_j is strongly stacked. $\square$

Since every strongly stacked frame is stacked, we obtain the following result.

Corollary 7.2. *Let A be a frame and $j \in NA$. If A is strongly stacked, then A_j is stacked.*

Proof. By the previous proposition, if A is strongly stacked, then A_j is strongly stacked and this implies that A_j is stacked. $\square$

8. RELACIÓN CON LA PROPIEDAD (H)

Si A es un marco Hausdorff, entonces A es T_1 . De esta manera, para poder utilizar el ejemplo de Paseka y Smarda basta que (H) implique apilado.

Proposition 8.1. *Si $A \in \text{Frm}$ satisface (H), entonces A es apilado.*

Proof. Si A satisface (H), entonces A es T_1 , es decir, para todo par HM se cumple que

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F,$$

es decir, $\mathcal{U}_* \circ [Q'] \circ \mathcal{U}(0) = w_F(0)$ y dado que $w_F \neq \text{tp}$, entonces $w_F(0) \neq 1$.

Por construcción se cumple que $v_F(0) \leq w_F(0)$ y así, debemos ver que se da la otra desigualdad para obtener que A es apilado. A manera de contradicción, supongamos que

$$w_F(0) \not\leq v_F(0).$$

Como $1 \neq w_F(0) \not\leq v_F(0)$, por (H), existe $c \in A$ tal que

$$c \not\leq w_F(0) \quad \text{y} \quad \neg c \not\leq v_F(0).$$

Si $\neg c \not\leq v_F(0)$, entonces $c \notin F$, pues en caso contrario, si $c \in F$, se cumple que

$$\neg c \leq f_F(0) = \bigvee \{v_a(0) \mid a \in F\} \leq v_F(0),$$

lo cual es una contradicción a (H).

Si $c \notin F$, entonces $c \in A \setminus F$ y por construcción existe $p \in M$ tal que $c \leq p$, donde

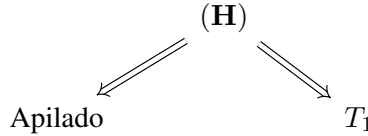
$$M = \{p \in A \setminus F \mid p \text{ es máximo}\}.$$

De aquí que $p \not\leq w_F(0)$. Luego,

$$w_F(0) = \bigwedge \{p \in M \mid 0 \leq p\} = \bigwedge M \leq p,$$

es decir, $p \not\leq w_F(0) \leq p$, lo cual es una contradicción. La contradicción viene de suponer que $w_F(0) \not\leq v_F(0)$. Por lo tanto $w_F(0) = v_F(0)$ y A es apilado. $\square$

Por el resultado anterior, y ya que T_1 y apilado son propiedades incomparables, tenemos la siguiente relación con la propiedad (H)



Usaremos este hecho para exhibir un ejemplo de lo siguiente.

Theorem 8.2. *Existen marcos T_1 y apilados que no son espaciales.*

Proof. Si consideramos $A \in \text{Frm}$ tal que A satisface **(H)**, podemos construir el siguiente marco

$$\tilde{A} = \{(a, b) \mid a \in A, b \in \mathcal{B}(A)\},$$

donde $\mathcal{B}(A) = \{b \in A \mid b = \neg \neg b\}$ y se conoce como la *booleanización* de A .

Además, si para $x \in A$, con x máximo, se cumple que $\neg x = 0$, entonces $\tilde{A}$ satisface **(H)**. En consecuencia, $\tilde{A}$ es un marco T_1 y apilado, pero no es espacial. $\square$

Para ver los detalles de la no espacialidad de $\tilde{A}$, se puede consultar [PŠ92].

9. MARCOS KC

Definition 9.1. Sea $A \in \text{Frm}$ decimos que A es:

- (1) *KC* si cada cociente compacto es cerrado.
- (2) *Hausdorff cerrado compacto* (o *KCH* de manera abreviada) si cada cociente compacto de A es cerrado y Hausdorff.

La Definición 9.1 nos dice que si $j \in NA$, entonces $\nabla(j) \in A^\wedge$ y $j = u_a$ para algún $a \in A$. De manera adicional, para 2) pedimos que A_j cumpla la propiedad **(H)**.

Proposition 9.2. Si $A \in \text{Frm}$ cumple *KC*, entonces A_j cumple *KC* para cada $j \in N(A)$.

Proof. Consideremos $k \in NA_j$ tal que $(A_j)_k$ es compacto. Como cualquier filtro abierto es admisible, tenemos que $\nabla(k) \in A_j^\wedge$ y por la Proposición 3.1 $j_* \nabla(K) \in A^\wedge$.

Sea $l = j_* \circ k \circ j^* \in NA$, entonces A_l es un cociente compacto de A y existe $a \in A$ tal que $l = u_a$. Así

$$\begin{array}{ccccccc} & & & l & & & \\ & & & \curvearrowright & & & \\ A & \xrightarrow{j^*} & A_j & \xrightarrow{k} & (A_j)_k & \xrightarrow{j_*} & A_j \subseteq A \end{array}$$

y $a \vee x = k(j(x))$. Por lo tanto, si $x = a$, $k(j(x)) = a$.

Necesitamos que $k = u_b$ para algún $b \in A_j$. Para $x \in A_j$ y $b = j(a)$

$$\begin{aligned} u_b(x) &= b \vee x = b \vee j(x) = j(j(a) \vee j(x)) \\ &= j(k(j(a)) \vee x) \\ &= j(u_a(x)) \\ &= j(k(x)) \\ &= k(x). \end{aligned}$$

Por lo tanto $u_b = k$. $\square$

Proposition 9.3. *Si A es un marco KC , entonces A es un marco T_1 .*

Proof. Sean $p \in \text{pt } A$ y $a \in A$ tales que $p \leq a \leq 1$. Consideremos

$$w_p(x) = \begin{cases} 1 & \text{si } x \not\leq p \\ p & \text{si } x \leq p \end{cases}$$

para $x \in A$. $P = \nabla(w_p) = \{x \in A \mid x \not\leq p\}$ es un filtro completamente primo (en particular, $P \in A^\wedge$). Como A es KC , entonces A_{w_p} es un cociente compacto cerrado. Así $u_p = w_p$ y

$$u_p(a) = a \quad \text{y} \quad w_p(a) = 1.$$

es decir, $a = 1$. Por lo tanto p es máximo. □

Nuestro objetivo es estudiar los núcleos que producen cociente compacto. De manera particular, estamos interesados en aquellos núcleos que producen cociente compacto y cerrado.

Para realizar parte de este estudio, definimos el siguiente subconjunto de A

$$uA = \{a \in A \mid A_{u_a} \text{ es un cociente compacto}\}.$$

Notemos que a todo $a \in uA$ le asignamos un subconjunto $\nabla(u_a) \in A^\wedge$, es decir, podemos definir la función

$$\alpha: uA \rightarrow A^\wedge.$$

Veamos que propiedades satisface dicha función.

Lemma 9.4. *Para $uA \subseteq A$ como antes, las siguientes afirmaciones se cumplen:*

- (1) *Si A es eficiente, entonces α es suprayectiva.*
- (2) *Si u_a es ajustado para todo $a \in uA$, entonces α es inyectiva.*
- (3) *Si A es eficiente y $a \in uA$, entonces α es biyectiva si y solo si u_a es ajustado.*

Proof.

- (1) Consideremos $F = \nabla(j) \in A^\wedge$ y v_F el núcleo ajustado asociado a F . Si A es eficiente, entonces $v_F = u_d$, donde $d = v_F(0) \in A$. Luego,

$$\alpha(d) = \nabla(u_d) = \nabla(v_F) = F.$$

Por lo tanto α es suprayectiva.

- (2) Consideremos $a, b \in uA$ tales que $\alpha(a) = \alpha(b)$, es decir, $\nabla(u_a) = \nabla(u_b)$. Por hipótesis, u_a, u_b son núcleos ajustados, es decir,

$$u_c \leq u_d \quad \text{y} \quad u_c \geq u_d.$$

Evaluando ambos núcleos en 0 obtenemos $c = d$ y por lo tanto α es inyectiva.

- (3) Sean A eficiente, $a \in uA$ y supongamos que α es un biyectiva. Consideremos $F = \nabla(u_a)$ y v_F el núcleo ajustado asociado F . Por la eficiencia, $v_F = u_d$, es decir,

$$\alpha(a) = \nabla(u_a) = \nabla(v_F) = \nabla(u_d) = \alpha(d)$$

donde $d = v_F(0)$.

Por hipótesis, α es biyectiva, entonces, por la inyectividad, $a = d$. Por lo tanto $v_F = u_a$ y u_a es ajustado.

Recíprocamente, como A es eficiente, entonces por 1), α es suprayectiva. Por hipótesis, u_a es ajustado y, por 2), α es inyectiva. Por lo tanto α es biyectiva.

□

Dado que todo marco KC es eficiente, podemos dar una versión del lema anterior involucrando los marcos KC .

Corollary 9.5. *Sean A un marco KC y $F \in A^\wedge$. α es biyectiva si y solo si para todo $j \in [v_F, w_F]$, j es ajustado.*

Proof. Supongamos que A es KC y sea $F \in A^\wedge$. Por definición, para todo $j \in [v_F, w_F]$, $j = u_a$ para algún $a \in A$. Dado que A es KC , entonces A es eficiente y por el lema anterior, α es biyectiva si y solo si para todo $j \in [v_F, w_F]$, j es ajustado. □

Proposition 9.6. *Sea $A \in \text{Frm}$. A es KC si y solo si A es eficiente y $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$.*

Proof.

$\Rightarrow$): Supongamos que A es KC . Trivialmente, KC implica eficiente y así, resta verificar que KC implica que $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ o equivalentemente, por propiedades del adjunto derecho,

$$v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}.$$

Si A es KC , entonces para cualquier $F \in A^\wedge$ se cumple que para todo $j \in [v_F, w_F]$, $j = u_a$ para algún $a \in A$. En particular,

$$v_F = u_d, \quad \text{y} \quad \mathcal{U}_* \circ v_\nabla \circ \mathcal{U} = u_a$$

donde $d, a \in A$.

Por construcción, siempre se cumple que $u_d \leq u_a$. Para ver la otra desigualdad verificamos que la función α es inyectiva. Notemos que $d \leq a$
 $\Leftarrow$): Supongamos que A es eficiente y $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ para toda terna HM (F, Q, ∇) . Primero, si A es eficiente, entonces A es T_1 . Además, si A es eficiente y $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$, entonces A es fuertemente apilado. De aquí que A es T_1 y fuertemente apilado. Lo anterior implica que $v_F = w_F$

y, por la eficiencia, $v_F = u_d$, donde $d = v_F(0)$. Por lo tanto, para todo $j \in [v_F, w_F]$, $j = u_d$, es decir, A es KC .

□

10. EJEMPLOS

10.1. **El marco** $[0, 1]$. Siguiendo la idea presentada en [Sim06b] (o, de manera alternativa, en [Sim04]), consideremos el marco linealmente ordenado

$$A = [0, 1].$$

En este caso se cumplen las siguientes observaciones.

(1) Para todo $a, b \in A$ la implicación $(a \succ b)$ está dada por

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

(2) Para $a, x \in A$, los núcleos distinguidos u_a, v_a, w_a están dados por

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases}, \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

(3) Si $F \subseteq A$ es un filtro, entonces F es siempre admisible y tiene la forma

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

para algún $a \in A$. En particular, $F \in A^\wedge$ si y solo si $F = (a, 1]$.

(4) Si $F \in A^\wedge$, el núcleo ajustado asociado v_F se calcula como

$$(16) \quad v_F(x) = l_m(x) = \begin{cases} 1 & \text{si } x > m, \\ x & \text{si } x \leq m. \end{cases}$$

donde $l_m = v_m \wedge w_m$ y $m = \bigwedge F$.

(5) Para $S = \text{pt } A$ se cumple que $S = [0, 1]$ y los abiertos en $\mathcal{O}S$ son de la forma

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

(6) El marco A no es T_1 , pues ningún $p \in S$ es máximo. En consecuencia, A no satisface ninguna propiedad de separación más fuerte en Frm .

(7) Para $F \in A^\wedge$ el conjunto compacto saturado asociado $Q \in \mathcal{Q}S$ es de la forma $Q = [0, m]$.

(8) Se cumple que $[Q'] = v_F$; por lo tanto, S es fuertemente apilado.

(9) Ya que $[0, x) \in \mathcal{O}S$, entonces $[x, 1] \in \mathcal{C}S$ y $Q = [0, m]$. Por lo tanto, los subconjuntos compactos no son cerrados; es decir, S no es empaquetado.

(10) El marco A no es eficiente.

Todo lo anterior resume las principales características del marco $[0, 1]$. En lo que sigue, realizaremos el análisis de las construcciones de parches tanto de A como de su espacio de puntos S .

Recordemos que para $A \in \text{Frm}$ y $S \in \text{Top}$ tenemos

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

donde $Pbase$ genera al marco de parches de A y $pbase$ genera la topología del espacio de parches de S . Por (16) tenemos que

$$Pbase = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1]\}.$$

Para $x \in A$ se cumple

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Analizamos por casos la relación entre x y m .

Caso 1: Si $x < m$, entonces $(u_a \wedge l_m)(x) = u_a(x) \wedge x = x$.

Caso 2: Si $x = m$, entonces $(u_a \wedge l_m)(x) = u_a(m) \wedge m = x$.

Caso 3: Si $x > m$, entonces $(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x)$.

En los primeros 2 casos el valor de a no afecta la evaluación. La dependencia esencial aparece cuando Además $u_a \wedge l_m = \text{id}$, es decir, los generadores básicos de la $m < x$, donde interviene la relación entre a y m .

Caso 3.1: Si $m < x < a$, entonces $(u_a \wedge l_m)(x) = a$.

Caso 3.2: Si $m < a \leq x$, entonces $(u_a \wedge l_m)(x) = x$.

De este análisis se deduce que, $u_a \wedge l_m = \text{id}$ si y solo si $m < a$. Por lo tanto, lo generadores no triviales de $Pbase$ corresponden exactamente a los pares (m, a) con $m < a$.

En consecuencia, el marco PA queda codificado por los intervalos abiertos de la forma (m, a) , donde $m \in \text{pt } A$ y $a \in A$.

Queremos identificar si en nuestro ejemplo, PA satisface alguna propiedad de separación. Para hacerlo, ocuparemos recordar información respecto a los parches.

Si $A \in \text{Frm}$ es un marco arbitrario y S su espacio de puntos, podemos construir el siguiente diagrama conmutativo que relaciona ambas construcciones de parches.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \hookrightarrow & NA \\ \mathcal{U}_A \downarrow & & \downarrow P(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & P\mathcal{O}S & \hookrightarrow & N\mathcal{O}S \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

donde $\mathcal{O}^f S$ es la topología de Skula (o topología frontal) del espacio S . En [SS06b] lo llaman el *diagrama del ensamble de parches* y más detalles sobre este también pueden ser encontrados en [Sex03].

En nuestro caso, $\mathcal{U}_A$ es un isomorfismo. Además, bajo $\mathcal{U}_A$, la construcción de parches resulta ser funtorial y, por lo tanto, $PA \cong P\mathcal{O}S$. De manera similar, $NA \cong N\mathcal{O}S$. Para describir explícitamente $P\mathcal{O}S$ y $N\mathcal{O}S$ utilizamos el siguiente resultado, que puede consultarse en [ÁBMZ20].

Proposition 10.1. *Sea S un conjunto parcialmente ordenado.*

- (1) Si S no tiene anticadenas infinitas, entonces NOS es espacial.
- (2) Si S es totalmente ordenado, entonces NOS es espacial.

Así, por la Proposición 10.1 tenemos que NOS es espacial y por lo tanto $NOS \cong \mathcal{O}^f S$. Juntando toda esta información el diagrama del ensamble de parches en nuestro caso particular queda como sigue.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong \downarrow & & \downarrow \cong & & \downarrow \cong \\ OS & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Por lo tanto, si queremos estudiar al marco PA podemos estudiar a la topología de parches $\mathcal{O}^p S$. Recordemos que

$$\text{pbase} = \{U \cap Q' \mid U \in OS, Q \in QS\}.$$

En nuestro caso,

$$\text{pbase} = \{[0, a) \cap [m, 1) \mid a \in [0, 1], m \in [0, 1)\} = \{(m, a)\}.$$

Como la topología generada por los intervalos (m, a) coincide con la topología usual en $[0, 1)$, el espacio $^p S$ es Hausdorff. En consecuencia, $\mathcal{O}^p S \cong PA$ satisface **(H)**, aun cuando A no es T_1 ni eficiente.

Lemma 10.2. Sea A un marco, si el morfismo $pU : pA \rightarrow pOS$ es un isomorfismo, entonces A es fuertemente apilado.

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q , thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we need to observe the action of pU on these two nuclei, in order to do that remember that we have a commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & pA & \xrightarrow{\iota} & NA \\ \mathcal{U}_A \downarrow & & \downarrow p(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ OS & \longrightarrow & pOS & \xrightarrow{\iota} & NOS \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

since $N(\mathcal{U})\iota = \iota p\mathcal{U}$, that is, $p\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to pA , then

$$p\mathcal{U}(v_F) = N(\mathcal{U})(v_F) = \bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee \{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $v_{\mathcal{U}F}$.

□

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